Pathway High Prep PS End Term Report

Team 37

December 7, 2025

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1 Introduction

Banks routinely encounter **sophisticated fraud patterns** that compromise operational integrity, including **identity masking, transaction structuring, coordinated mule activity, and rapid cross-border layering**. These behaviours exploit latency, stale data, and weak entity resolution, making them difficult to detect within traditional batch-oriented systems. In such settings, incomplete KYC information, outdated sanctions feeds, and unverified media signals widen the exposure window and create systematic blind spots. Batch ETL amplifies this weakness by delaying re-screening until the next scheduled cycle, forcing expensive backfills and allowing malicious accounts to operate undetected for prolonged periods.

Our Product ('***Vigil360***') addresses these challenges by functioning as a **streaming, compliance-grade** intelligence platform that continuously fuses **onboarding KYC, sanctions and watchlists, authenticated adverse media, and Customer-360 activity signals** into a unified, event-driven entity state. Built on *Pathway* with high-velocity ingestion via **Kafka Connector**, each incoming event like new KYC record, OFAC/PEP [1] update, classified media item, or transaction incrementally mutates the materialized view and triggers immediate, deterministic re-scoring. High-confidence deterministic signals (e.g., exact sanctions matches) override probabilistic components such as media authenticity, fuzzy-identity match signals, anomaly detectors, and supervised behavioural models. These are combined into a calibrated *Risk Propensity Score* $\rho(u, t)$ and aggregated into ρ_{360} across **rolling windows** to support operational decision-making.

The system is engineered around **bank-centric data sovereignty**, where each institution maintains its own **isolated state** and inference flow. The ingestion layer supports configurable retention and TTL for feature stores, and **Pathway's Persistence** checkpointing ensures recovery without full rescans. Each decision produces an immutable audit artefact containing the feature snapshot, model versions, triggered rules, likelihood ratios, and downstream outcomes. Designed for deployment across both **retail and mid-market banking** environments, *Vigil360* enables **continuous KYC, explainable fraud detection, and adaptive risk scoring** while supporting **human-in-the-loop workflows**. Core evaluation metrics include **precision, recall, AoC, detection latency, false-positive rate, and broadcast latency**, which collectively guide drift monitoring, recalibration, and continuous system hardening. The Github[1]. repository can be found at (`Link`) which is currently private and would be made public on the day of presentation (as per guidelines).

2 Uniqueness

Our architecture introduces a **real-time, multi-agent circular feedback loop** that updates customer risk whenever new signals arrive. A central Checker Agent fuses inputs from KYC, sanctions, adverse media, social media, and Customer 360 streams into an immediate toxicity recalculation. This design enables continuous risk adjustment instead of batch or linear processing. We use a ρ_0–ρ_{360} risk score framework that converts heterogeneous signals into a **unified, explainable score**. A Consolidator and feedback mechanism incorporate **investigator decisions and suspicious activity** back into the risk model. Our fraud algorithm achieves significantly faster processing by leveraging Pathway's high-throughput streaming engine and optimized feature pipelines. Together, this architecture provides instant, adaptive, and computationally efficient risk intelligence beyond the capabilities of conventional fraud detection systems.

[1] `https://github.com/AndrewGG-TheFirst/Pathway_Team_37_High_Prep_InterIIT14`

3 Problem Understanding & Use-Case Validation

3.1 Requirement of a Streaming-native Setup

Real-time KYC, sanctions screening and transaction surveillance are inherently time-sensitive and adversarial; hence *Vigil360* is designed as a streaming-native system (live tables, real-time indexes, continuously updated metrics) rather than a batch ETL pipeline. Streaming primitives are required for three concrete technical reasons: (1) **low-latency detection and mitigation** - entity state must be recomputed and acted on within seconds of a new event (KYC parse, sanctions update, media mention or transaction) to prevent loss from rapid fraud typologies such as layering and mule networks; (2) **incremental stateful computation** - per-entity live state tables with event-time watermarks, exactly-once semantics and checkpointed state avoid expensive full re-scans while retaining correct aggregates across out-of-order inputs; and (3) **low-latency retrieval** - live vector indices and materialized feature-views enable fast semantic retrieval and explanation like nearest adverse-media evidence for analyst workflows. Taken together, these streaming capabilities are necessary to shrink the exposure window, preserve auditability (immutable decision artefacts), and operate safely in production under adversarial pressure.

3.2 Evidence of Production relevance and Data availability

The requirement for **real-time, robust KYC and AML tooling** is supported by industry analysis and practitioner feedback. McKinsey [2]argues that banks must accelerate enterprise AI adoption, modernize data platforms, and embed continuous, evidence-driven pipelines to remain competitive—recommendations that align directly with *Vigil360*'s streaming-first design. Moody's highlights [3] that generative AI can enable interactive, investigator-centric screening that reduces false positives and speeds enhanced due diligence, while stressing the need for governance and calibration. This supports *Vigil360*'s LLM-assisted fusion layer and strong audit logging. From a regulatory and systemic perspective, continuous monitoring is indispensable because banks hold monetary liabilities and must demonstrate ongoing due diligence against sanctions, PEP exposure and adverse media to avoid systemic and legal risk. After talking with few Industry level experts, we reported the following. Senior KYC/AML operators in Indian retail banking (*Axis Bank* KYC hubs) reported throughput constraints (**100–150 forms/day per hub** with some cases **taking up to 70 days to resolve**), and compliance analysts at multinational custody/asset banks (BNY / ex-Goldman practitioners) indicated that conventional AML detectors are **slow to surface emerging patterns** - both observations validate the operational benefit of **streaming augmentation**. Finally, sufficient data exists for development and evaluation- we combine institutionally feasible sources (watchlists and OpenSanctions [4]), public research corpora (Kaggle AML/transaction datasets and curated loan/credit datasets), and synthesized, adversarially seeded examples to build a representative testbed that supports offline model training and online rolling-window evaluation. Collectively, these industry signals, regulatory constraints, practitioner insights, and accessible datasets justify *Vigil360* as a realistic, production-worthy streaming solution for modern banking compliance.

4 Solution Overview

4.1 Design & Streaming Use

Vigil360 unifies **deterministic rules, supervised models and LLM-assisted fusion** in a *Pathway*-native, streaming-first platform- parsed KYC, sanctions/watchlist hits and adverse-media (with

fuzzy-name and **image-match signals**) stream via **Kafka Connector** into *Pathway* streaming tables, with **Postgres+Debezium** [5] [6] providing durable change-capture and *Pathway Persistence* enabling **warm restarts** and exactly-once state. Incoming events are normalized (schema alignment, entity resolution, fuzzy matching) to produce live features (authenticity, anomaly, relevance) consumed by *Pathway* agents (Adverse Media, Sanctions, Consolidator) that update (ρ_0) and materialize low-latency indices where **live vector indices** power **fast RAG retrievals**. Transactional streams drive the behavioural risk score ρ_{360}, computed through **rolling-window aggregates** and **anomaly-aware scoring**. When ρ_{360} crosses a threshold, the **Pathway-native MCP server** selects verification tools (sanctions checks, adverse-media scrapers, RAG lookups) to validate the behavioural explanation and emits a structured validation signal. This validation interacts with the post-aggregation logic to update ρ_0 or trigger deterministic enforcement. A **scheduled live-RAG agent** continuously re-embeds new articles, updates vector indices incrementally, and issues deltas that cause targeted recomputation of ρ_0.

4.2 Agent Flow

Decisions follow an event-driven planner→tools→action flow on *Pathway* MCP with a live RAG agent. On initial KYC parse the adverse-media and sanctions agents run scrapers, fuzzy-name and image-search matching, and produce candidate evidence which a deterministic consolidator fuses into (ρ_0) and persists to streaming tables; a scheduled RAG agent continuously re-embeds new evidence and issues deltas that trigger (ρ_0) recomputation. Transactional monitoring computes rolling-window behavioural risk (ρ_{360}); when (ρ_{360}) exceeds a threshold the engine emits the score and rationale to the MCP planner, which selects validation tools (scrapers, sanctions resolvers, RAG lookups), aggregates validated signals into a coherence value, and forwards the result to post-aggregation logic to update (ρ_0) or execute deterministic actions. Hard policy rules short-circuit for immediate blocks, **ambiguous/high-impact cases are routed to human analysts** with packaged evidence bundles (feature snapshot, fired rules, LLM explanation, model versions), and every agent action emits **immutable audit records** for compliance.

5 System Architecture

5.1 Overview

Vigil360 is implemented as an agentic, streaming-first pipeline (fig:1) built entirely on *Pathway's* primitives. The system is organized into logical layers which are **ingestion and capture, canonicalization and enrichment, agentic inference and retrieval, post-aggregation and actioning, and storage/UI/broadcast**. *Pathway* provides end-to-end support for connectors, streaming tables, MCP orchestration, live RAG agents, simple agents (LLM wrappers), UDFs, live vector indexing, and persistence (warm-restart checkpointing). All core components and integrations are powered by *Pathway* and its native wrappers and no external orchestration or streaming runtime is required. This consolidation enables a single, auditable system of record where features, indices, models and decisions remain synchronized and actionable in real time.

5.1.1 Pathway Modules Usage

All modules below are executed inside *Pathway* (using its native adapters/wrappers):

- **MCP servers** [7] (planner / orchestrator for validation workflows and tool selection).

- **Live RAG Agent** (scheduled evidence re-checks, incremental embedding inserts, live vector indices).
- **Simple Agents / LLM XPack** (LLM-assisted fusion, reasoning, and explainability).
- **Persistence** (checkpointed streaming state for warm restarts).
- **Connectors & UDFs** (native *Pathway* connectors for IO, Kafka, Postgres, image/document ingestion; user UDFs for phonetic/fuzzy matching and image-search wrappers).
- **Live indexing** of documents

5.1.2 Adverse Media Classification and Triage Tool

A tool scrapes news sites and social media platforms to identify and cluster information concerning an individual when they appear in coverage. **Fuzzy matching** is employed to reduce false positives arising from name similarities, ensuring that all linked information corresponds accurately to the intended individual. **The authenticity of each media source is assessed** to verify that the information is credible and not biased towards any segment of the population. Guardrails [8] further eliminate content unsuitable for automated classification.

The adverse-media score for customer c at time t is computed as:

$$M(c,t) \;=\; \max_{a \in A(c,t)} \Big(\mathrm{auth}(a) \cdot \mathrm{rel}(a,c) \cdot \mathrm{match}(a,c)\Big),$$

where $A(c,t)$ denotes the set of candidate articles, $\mathrm{auth}(\cdot)$ is the authenticity weight of each source, $\mathrm{rel}(\cdot,\cdot)$ measures semantic relevance, and $\mathrm{match}(\cdot,\cdot)$ is the fuzzy-identity match score.

5.1.3 Sanctions Reconciliation Tool

Names are matched against dynamic regulatory datasets, including **Politically Exposed Persons (PEP) lists, national sanctions databases, and arrest registries**. This tool operates over high-velocity feeds and generating summarized matches for compliance verification and downstream processing.

The sanctions score for customer c at time t is defined as:

$$S(c,t) \;=\; 1 \;-\; \prod_{i \in L} \big(1 - \mathrm{match}_i(c,t)\big),$$

where L indexes all monitored lists and $\mathrm{match}_i(c,t) \in [0,1]$ represents the normalized probability of a match on list i.

5.1.4 Checker / Consolidator Agent

This agent consolidates the outputs of the Adverse Media and Sanctions agents to compute an initial registration RPS score, denoted as $\rho_0(c)$. The score represents the **baseline Risk Propensity** of a new customer at onboarding, integrating both structured sanctions evidence and unstructured media signals. A language model–driven reasoning layer interprets the qualitative context extracted from media and sanctions summaries, assigning a unified probabilistic risk score.

5.1.5 Live RAG Agent with adaptive scheduler (AIMD) and similarity trigger

Let s_t denote the top-k similarity index returned by the live RAG agent [9] for recently ingested evidence. Define threshold θ_{sim} below which freshness/coverage is considered degraded. The scheduler maintains a scheduling window W_t (polling frequency or re-check budget) updated with an Additive Increase Multiplicative Decrease (AIMD) rule:

$$W_{t+1} = \begin{cases} \min(W_{\max},\, W_t + \alpha), & s_t \geq \theta_{\text{sim}} \quad \text{(no deficit)}, \\ \max(W_{\min},\, \beta W_t), & s_t < \theta_{\text{sim}} \quad \text{(deficit detected)}, \end{cases}$$

with $\alpha > 0$ (additive increase) and $0 < \beta < 1$ (multiplicative back-off). A re-index or immediate evidence re-fetch is triggered when $s_t < \theta_{\text{reindex}} \leq \theta_{\text{sim}}$.

5.1.6 Risk Propensity Score

The Risk Propensity Score(RPS) combines three complementary sources of fraud risk: (i) **Supervised machine learning probability**, (ii) **Anomaly-based probability**, and (iii) **Bayesian rule–evidence** derived from expert behavioural rules. The supervised model (CatBoost [10]) outputs a calibrated fraud probability $p_{\text{ML}} = P(\text{fraud} \mid x)$, while the anomaly model (Isolation Forest [11]) produces a normalized anomaly score $p_{\text{anom}} \in [0, 1]$. For each behavioural rule r_i, we compute a likelihood ratio

$$LR_i = \frac{P(r_i = 1 \mid \text{fraud})}{P(r_i = 1 \mid \text{legit})},$$

which updates the prior fraud probability π through Bayesian odds. Let $O_0 = \frac{\pi}{1-\pi}$ denote prior odds and let $\mathcal{R}$ denote the set of triggered rules. The posterior evidence score is

$$e = \frac{O_0 \prod_{i \in \mathcal{R}} LR_i}{1 + O_0 \prod_{i \in \mathcal{R}} LR_i}.$$

The final RPS is produced by a **logistic-regression fusion model** that combines the logit-transformed supervised and anomaly scores with the Bayesian evidence:

$$\rho_{360} = P(\text{fraud} \mid p_{\text{ML}}, p_{\text{anom}}, e) = \sigma(w_1 \operatorname{logit}(p_{\text{ML}}) + w_2 \operatorname{logit}(p_{\text{anom}}) + w_3 e + b)\,,$$

where $\sigma(\cdot)$ is the logistic function. This fusion provides a unified probabilistic estimate of fraud by integrating supervised learning, anomaly detection, and rule-based reasoning.

5.1.7 Transaction alarm, MCP validation and Comment

When rolling-window behavioural monitoring produces $\rho_{360}(c, t)$ and $\rho_{360} \geq \tau$ (alert threshold), the MCP planner selects a set of validation tools $\mathcal{T} = \{T_j\}$ (scrapers and sanction resolvers). Each tool returns a match confidence $m_j \in [0, 1]$ and it sets a validation comment.

5.1.8 Post-aggregation Logic

We combine the behavioural alarm $y = \rho_{360}$ and the current ρ_0 score into a coherence value

$$z \;=\; 1 - (1 - x)(1 - y) \;=\; x + y - xy,$$

interpretable as the probability that at least one of the signals indicates risk.

6 Results and Metrics

Due to the **sensitive nature** of data related to KYC forms and transactions datas, we synthesized our own dataset (Appendix Table 2) and ran known products and compared our product on the basis of standard metrics whose definitions are provided in appendix.

Table 1: Core Performance Metrics Comparison

Metric	***Vigil360(Ours)***	**Lucinity**	**Moody's AI**	**Akira AI**	**UiPath**
Latency(R0)	68s	Private	Private	Private	Private
Latency(R360)	0.5s	Private	Private	Private	Private
Latency(MCP)	100s	Private	Private	Private	Private
Google API Relevance Precision	0.94	No	No	No	No
MCP Response Accuracy	100%	No	Yes	Yes	No
MCP Source Verification	96%	No	No	No	No
RAG Accuracy	0.92	0	0	0	0
AUC-ROC (Risk Model)	0.936	Private	Private	Private	Private

7 Implementation Details and Production Bar

The pipeline ingests KYC entities and transactions into *Pathway* streaming tables (Kafka + Postgres/Debezium sources), maintains live risk indexes via group-by/reduce patterns, and combines deterministic rules, ML scoring and MCP-orchestrated agent calls for end-to-end coverage. Onboarding KYC is **OCR-parsed** via an **AWS S3 Bucket** and enriched (OpenSanctions, Web analysis), filtered through guarded LLMs, and persisted as **append-only JSONL logs** plus latest-per-entity snapshots exported to **Kafka/Postgres staging buffer**. Transaction stream feeds a rolling-window RPS engine that fuses supervised probability and anomaly signals into ρ_{360}; thresholded alarms spawn MCP validation workflows (selected OFAC/PEP/news tools) which return match confidences used to update ρ_0. A watchdog **adaptive live scheduler** re-processes snapshots, computes **cosine similarity** on debug web outputs to detect content drift, and triggers RAG re-scoring and deterministic fusion when needed.

Production hardening emphasizes durability, observability and safe fallbacks. Kafka provides durable event logs/consumer groups and offset recovery while *Pathway* recomputes derived tables from persisted checkpoints; Debezium streams allow incremental recovery without full reprocessing. Writes use safe UDFs and atomic JSONL swaps; MCP calls have bounded timeouts and persistence; rule-based RPS or prior scores act as fallbacks when ML/RAG paths fail. Observability includes per-stage JSONL traces, structured logs with correlation IDs, scheduler/DB metrics and report statistics. Operational guardrails (LLM content filters, strict RAG JSON parsers, schema typing, typed UDF tests, subprocess timeouts), environment-driven configuration, single-command entrypoints and role-specific scripts complete the deployment surface for predictable, auditable operation.

8 Challenges Faced

Development exposed a set of recurring operational challenges common to streaming-native compliance systems. **LLM-wrapper latency** and timeout volatility (exacerbated by free-tier keys), **OCR/backpressure** issues in PDF ingestion, unstable **third-party API latency** and rate limits, persistence/checkpointing failures during early integration, **opaque error traces**, entity-resolution

edge cases (transliterations, aliases, multilingual forms), schema drift and scarce labeled fraud examples. We mitigated these with engineering controls. LLM-response caching, batched/asynchronous inference and model-fallbacks (local light models for low-risk paths, paid keys for critical flows); **RAG resilience** via **chunk tuning**, incremental inserts and periodic compaction; ingestion decoupling through a **AWS staging layer**; API resilience using persistence; persistence stability via pinned dependencies, containerized runtimes, stream-validated integration tests and automated snapshot/restore; observability improvements with **structured JSON logs** and correlation IDs; and hardened entity resolution using phonetic encoders. To address drift and label scarcity we instituted adversarial data seeding, randomized audits, shadow/canary deployments and local service simulators, ensuring graceful degradation, reproducibility and auditable decision trails under production conditions.

9 Lessons Learned

Streaming ML combined with agentic AI maps naturally to the financial domain: live state, incremental features, and callable agents enable low-latency detection, clear audit trails, and safe human-in-the-loop escalation for compliance workflows. Implementing the pipeline on *Pathway* demonstrated that its native connectors, checkpointed streaming tables, MCP orchestration and live RAG agents materially simplify productionizing continuous inference, incremental model updates, and semantic evidence retrieval, reducing engineering friction and improving end-to-end responsiveness. These lessons drove improvements in resilient ingestion, index-maintenance, and lightweight online learning patterns that make *Vigil360* operable at bank scale.

10 Future Work

A promising enhancement is the introduction of a **post-Transformer (BDH-style) [12] reasoning layer** that replaces the current static post-aggregation rule with a learnable fusion module. In this formulation, the transaction-based score ρ_{360}, validation confidence values, and auxiliary temporal or behavioral features $\mathbf{h}$ are passed through a calibrated mapping $z_\phi = g_\phi(\rho_{360}, v, \mathbf{h})$, where g_ϕ is a lightweight post-Transformer component designed for continual adaptation. This learned coherence score z_ϕ can then be used to update the overall risk estimate, allowing the system to capture **higher-order interactions** and **evolving fraud patterns** more effectively than a static heuristic. Because such a module supports uncertainty calibration and incremental updating, it can refine risk assessments in real time while remaining compatible with existing deterministic policies. As the architecture matures, this extension naturally broadens *Vigil360* into a more general regulatory-compliance engine capable of adaptive screening, automated evidence synthesis, and richer cross-signal reasoning.

11 Conclusion

Vigil360 demonstrates the feasibility of an agentic, streaming-ML approach to production-grade KYC and fraud detection: a *Pathway*-native pipeline that fuses parsed KYC, sanctions/watchlists, authenticated adverse media, and Customer-360 behavioural signals into an auditable, entity-centric state. Preliminary development and scenario testing show that live tables, incremental indices and agent orchestration (MCP + Live RAG + LLM-assisted consolidators) enable low-latency re-scoring, explainable evidence bundles for analysts, and robust persistence for warm restarts. Key delivered outcomes include a working Streaming KYC pipeline, implemented adverse-media authenticity scoring, sanctions reconciliation, an initial consolidator producing ρ_0, a transactions monitoring prototype that emits ρ_{360}, and end-to-end UI and case artefacts for audit and review.

A Appendix

A.1 Architecture

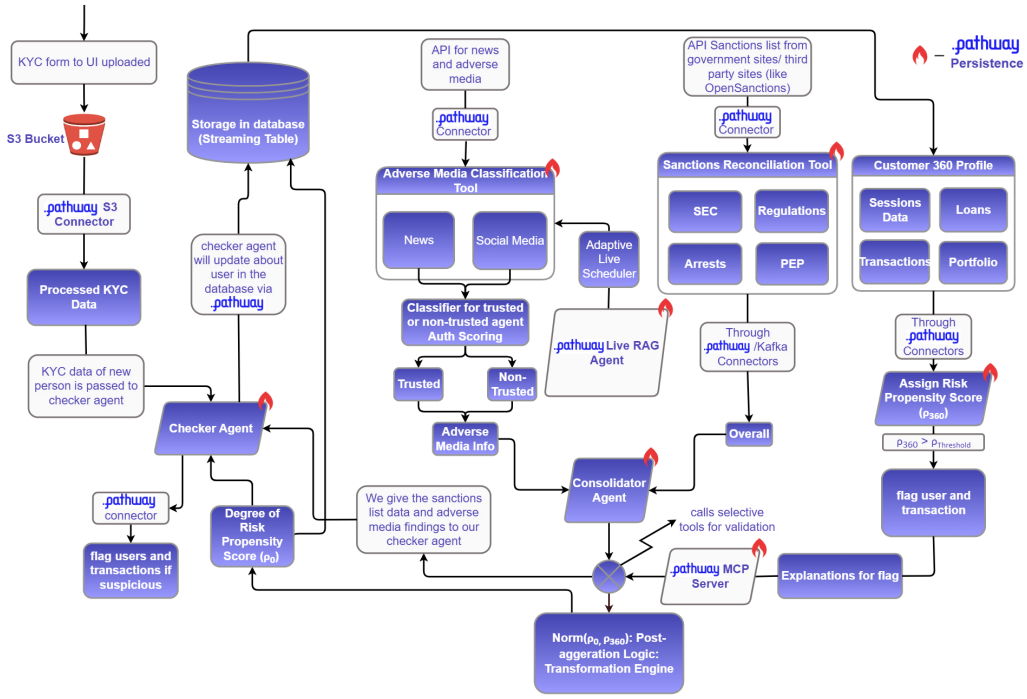

Figure 1: Architecture

A.2 Dataset

Datasets

Dataset	Total Records	Fraud Cases	Non-Fraud	Fraud Ratio (%)
AMLSim	540,175	1,719	538,456	0.318
SAML-D	39,289	49	39,240	0.125
Overall	**579,464**	**1,768**	**577,696**	**0.305**

Table 2: Summary of Dataset characteristics

The unified dataset used in our experiments consists of 579,464 transactions merged from two primary sources: AMLSim and SAML-D. AMLSim contributes the majority of records (540,175 transactions) and contains 1,719 labeled fraud cases, corresponding to a fraud ratio of 0.318%. SAML-D, while smaller with 39,289 entries, includes 49 fraud cases (0.125%), offering additional variation in transaction structure and fraud typologies. Across both datasets, fraud remains extremely sparse—only 1,768 cases in total, representing 0.305% of all transactions—mirroring real-world financial crime distributions where illicit activity is rare and heavily concealed.

A.3 Metrics Formula

$$\text{Relevance Precision} = \frac{\#\ \text{Relevant Articles}}{\#\ \text{Retrieved Articles}}$$

$$\text{Accuracy} = \frac{\#\ \text{Correct Responses}}{\text{Total Responses}}$$

$$\text{Source Verification} = 100 \times \frac{\#\ \text{Answers With Valid Source Links}}{\#\ \text{Total Answers}}$$

A.4 User Interface Design

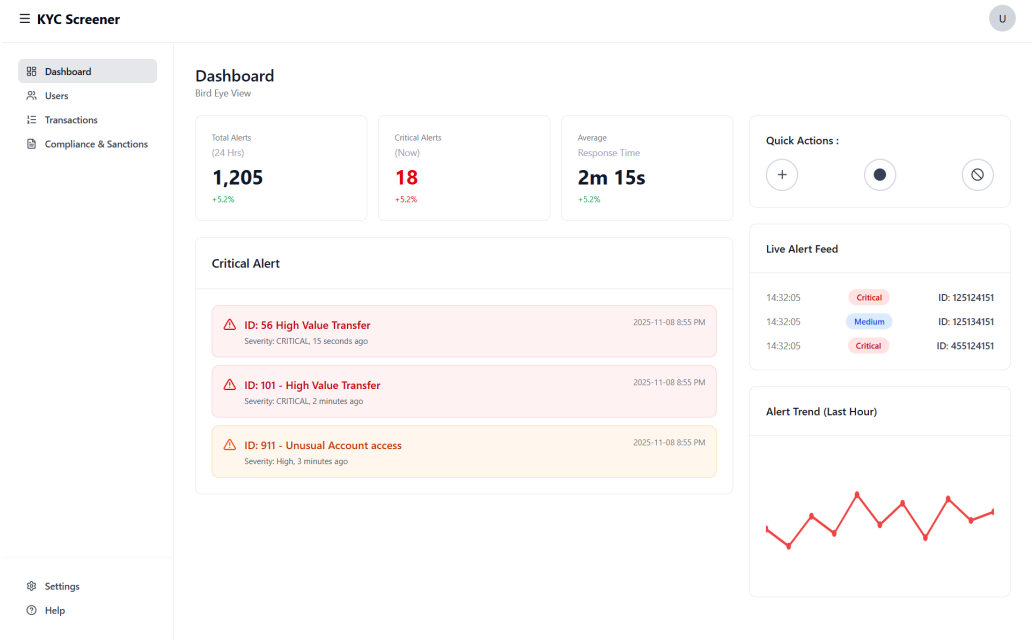

Figure 2: Dashboard

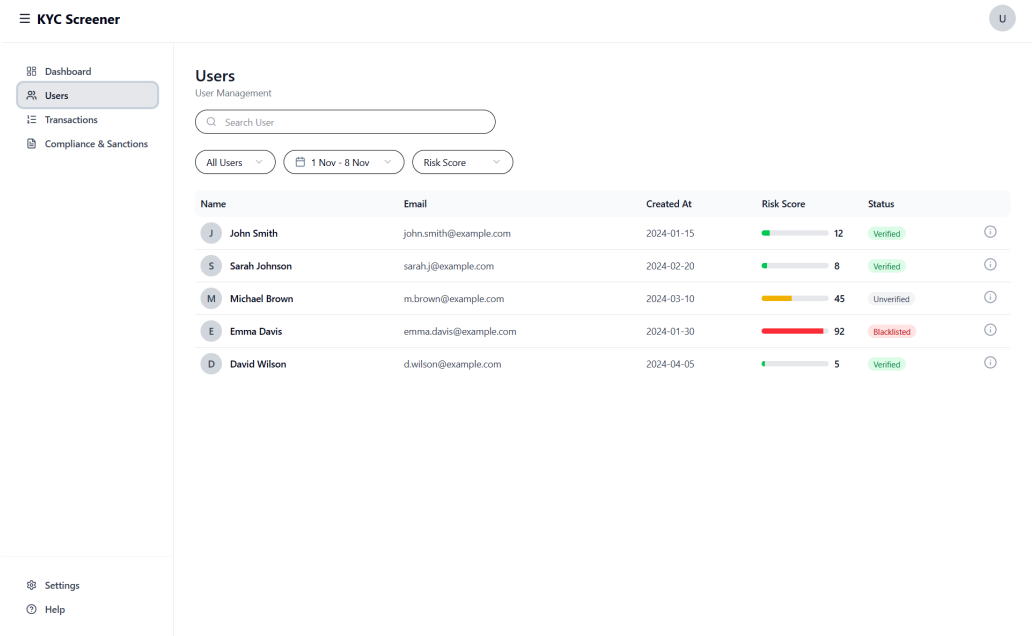

Figure 3: Users Page

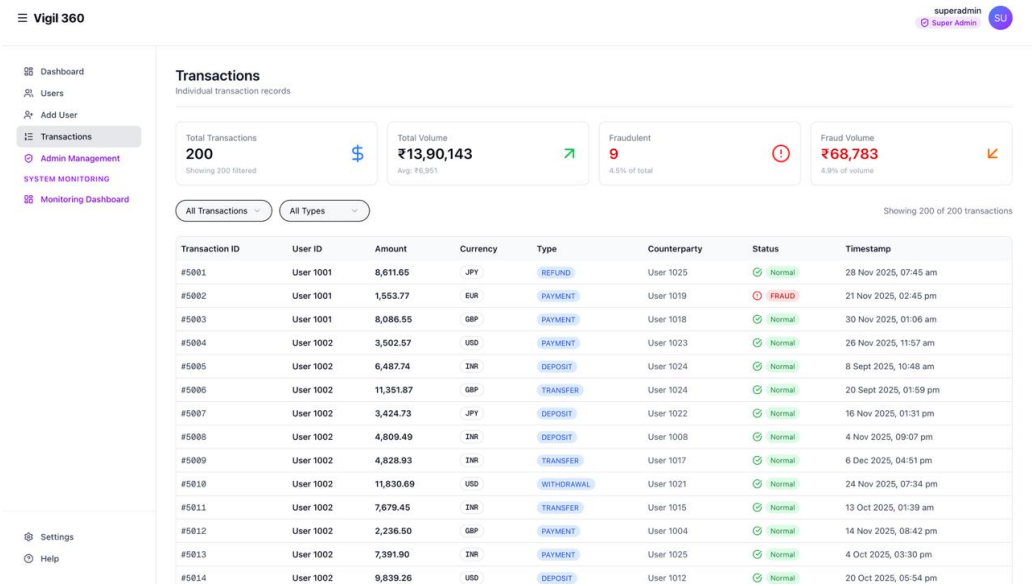

Figure 4: Transactions Page

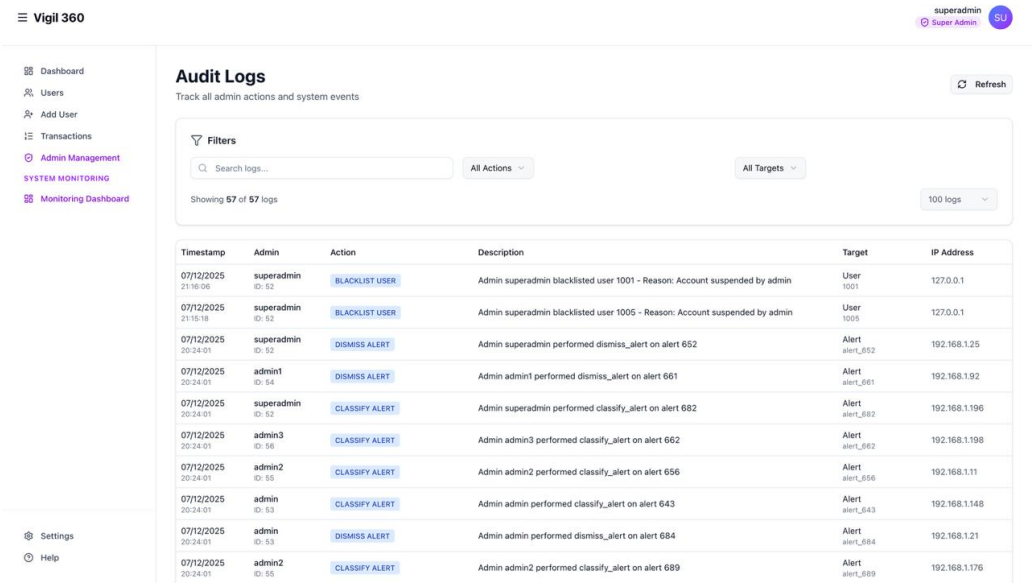

Figure 5: Audit Logs Page

A.5 Outputs

A.5.1 Web Scraper

streaming > out > {} web_analysis_debug.jsonl

```
1 {"entity_id":"1","name":"Nirav Modi","os_entity_name":null,"os_score":1.0,"web_prompt":"Web findings for **nirav modi**:\n\n--- Article 1
---\nTitle: Nirav Modi case: Swiss to get ED request\nURL: https://www.thehindu.com/news/national/nirav-modi-case-swiss-to-get-ed-request/
article25172273.ece\nAuthenticity Score: 0.90 (Reasoning: The article is from a well-established and reputable news source, 'The Hindu,' which
has a strong domain reputation and history. The content is well-structured, includes specific details about the Enforcement Directorate's
actions, and mentions legal processes, which adds to its credibility. The high Content Signals Score (0.8) indicates good journalistic
practices. The domain's whitelisted status and verified history further support the authenticity of the article.)\n(History: True,
AlienVaultHint: Whitelisted domain thehindu.com, OTX pulses: 0, OTX tags: )\nSnippet: The Enforcement Directorate is sending a judicial request
to Switzerland seeking confirmation of attachment of assets in connection with the Punjab National Ban...\n\n--- Article 2 ---\nTitle: Fugitive
Indian billionaire Nirav Modi last seen in Britain in March\nURL: https://www.theguardian.com/world/2018/jun/19/
fugitive-indian-billionaire-nirav-modi-last-seen-in-britain-in-march\nAuthenticity Score: 0.90 (Reasoning: The article is from The Guardian, a
well-established and reputable news source with a strong domain reputation. The content includes specific details such as the involvement of
the Central Bureau of Investigation (CBI) and mentions of Nirav Modi's net worth according to Forbes, which adds credibility. The article is
well-structured and provides clear facts. However, the domain age check indicates that the domain does not have a history, which slightly
reduces the score. The high content signals score and the prominent Alexa and Akamai ranks contribute positively to the overall authenticity
score.)\n(History: False, AlienVaultHint: Alexa rank: ■#172, OTX pulses: 0, OTX tags: )\nSnippet: The fugitive billionaire jeweller Nirav Modi
was last seen in Britain in March, Indian federal investigators have said.\n\nThe Central Bureau of Investigation (CB...\n\n--- Article 3
---\nTitle: How Mehul Choksi and Nirav Modi pulled off multi\nURL: https://timesofindia.indiatimes.com/business/india-business/
how-mehul-choksi-and-nirav-modi-pulled-off-multi-billion-dollar-pnb-scam/articleshow/120272112.cms\nAuthenticity Score: 0.85 (Reasoning: The
article is published on the Times of India, a well-established news publisher with a strong reputation (Alexa Rank ■#134, Akamai Rank □#4091).
The domain has a long history (Wayback Machine confirms domain age), which adds to its credibility. The article mentions specific details such
as the involvement of Indian probe agencies, the PNB scam, and the arrest of Mehul Choksi, which are verifiable facts. The Content Signals
Score of 0.6 indicates decent journalistic hygiene. However, the abrupt cut-off in the text and lack of additional context or sources slightly
reduces the score.)\n(History: True, AlienVaultHint: Alexa rank: ■#134, OTX pulses: 0, OTX tags: )\nSnippet: Mehul Choksi, a key figure in the
Rs 13,000 crore PNB fraud, has been arrested in Belgium following an extradition request by Indian probe agencies. This action...","citations":
["- [1] https://www.thehindu.com/news/national/nirav-modi-case-swiss-to-get-ed-request/article25172273.ece","- [2] https://www.theguardian.com/
world/2018/jun/19/fugitive-indian-billionaire-nirav-modi-last-seen-in-britain-in-march","- [3] https://timesofindia.indiatimes.com/business/
india-business/how-mehul-choksi-and-nirav-modi-pulled-off-multi-billion-dollar-pnb-scam/articleshow/120272112.cms"],"diff":1,
"time":1762899447766}
2 {"entity_id":"2","name":"Mehul Choksi","os_entity_name":null,"os_score":0.9,"web_prompt":"Web findings for **mehul choksi**:\n\n--- Article 1
---\nTitle: How Mehul Choksi and Nirav Modi pulled off multi\nURL: https://timesofindia.indiatimes.com/business/india-business/
how-mehul-choksi-and-nirav-modi-pulled-off-multi-billion-dollar-pnb-scam/articleshow/120272112.cms\nAuthenticity Score: 0.85 (Reasoning: The
article is from a well-established news source, 'The Times of India,' which has a strong reputation and a low Alexa rank (#134), indicating
high traffic and credibility. The domain has a long history, further supporting its reliability. The content signals score of 0.6 suggests
decent journalistic hygiene, though it could be improved with more detailed sourcing and author information. The article discusses a
significant financial scam involving high-profile figures and mentions extradition requests by Indian probe agencies, which adds to its
authenticity. However, the abrupt ending and lack of additional context slightly reduce the score.)\n(History: True, AlienVaultHint: Alexa
rank: ■#134, OTX pulses: 0, OTX tags: )\nSnippet: Mehul Choksi, a key figure in the Rs 13,000 crore PNB fraud, has been arrested in Belgium
following an extradition request by Indian probe agencies. This action...\n\n--- Article 2 ---\nTitle: Punjab National Bank Scam\nURL: https://
en.wikipedia.org/wiki/Punjab_National_Bank_Scam\nAuthenticity Score: 0.60 (Reasoning: The article URL is from Wikipedia, which is a well-known
and established platform with a long history, contributing positively to the authenticity score. However, Wikipedia articles are user-edited
and may not always reflect the highest journalistic standards, which slightly reduces the score. The content signals score is low, indicating a
lack of author attribution, date, and structured data, which further lowers the score. The domain's reputation and age are strong indicators of
reliability, but the lack of direct journalistic rigor in the content keeps the score from being higher.)\n(History: True, AlienVaultHint:
Alexa rank: #13, OTX pulses: 0, OTX tags: )\nSnippet: 2018 scandal in India\n\nThe Punjab National Bank Fraud Case relates to a fraudulent
letter of undertaking worth ₹12,000 crore (US$1.4 billion) issued by the Punj...","citations":["- [1] https://timesofindia.indiatimes.com/
business/india-business/how-mehul-choksi-and-nirav-modi-pulled-off-multi-billion-dollar-pnb-scam/articleshow/120272112.cms","- [2] https://en.
wikipedia.org/wiki/Punjab_National_Bank_Scam"],"diff":1,"time":1762899447766}
```


Figure 6: Web Analysis Results for few inputs

A.5.2 Initial Risk Propensity Score

streaming > out > {} reports.jsonl

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1 {"entity_id":"1","name":"Nirav Modi","os_entity_name":null,"os_score":1.0,"risk_json":{"risk_score":100,
"risk_classification":"CRITICAL","match_found":true},"summary":"LLM not configured/failed; deterministic risk
classification used.","citations":["- [1] https://www.thehindu.com/news/national/
nirav-modi-case-swiss-to-get-ed-request/article25172273.ece","- [2] https://www.theguardian.com/world/2018/jun/
19/fugitive-indian-billionaire-nirav-modi-last-seen-in-britain-in-march","- [3] https://timesofindia.
indiatimes.com/business/india-business/
how-mehul-choksi-and-nirav-modi-pulled-off-multi-billion-dollar-pnb-scam/articleshow/120272112.cms"],
"timestamp":1762899447.694274,"diff":1,"time":1762899447766}
2 {"entity_id":"2","name":"Mehul Choksi","os_entity_name":null,"os_score":0.9,"risk_json":{"risk_score":90,
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classification used.","citations":["- [1] https://timesofindia.indiatimes.com/business/india-business/
how-mehul-choksi-and-nirav-modi-pulled-off-multi-billion-dollar-pnb-scam/articleshow/120272112.cms","- [2]
https://en.wikipedia.org/wiki/Punjab_National_Bank_Scam"],"timestamp":1762899447.694274,"diff":1,
"time":1762899447766}
```


Figure 7: Initial RPS calculated ("risk_score") from Web Scraped Results

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